

Regularization Techniques for Image Reconstruction: beyond Tikhonov

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Image Reconstruction

Forward Problem

$$f = K\hat{u} + n$$

- $\Omega \subset \mathbb{R}^2$ is an open, bounded domain.
- $\hat{u} \in L^2(\Omega)$ is the true image.
- $f \in L^2(\Omega)$ is the data (or measurements).
- $K \in \mathcal{L}(L^2(\Omega))$ is the forward operator, e.g., blurring operator, Radon transform, Fourier transform, framelets, etc.
- n represents the noise. Here, we consider additive white Gaussian noise with mean 0 and variance σ^2 .

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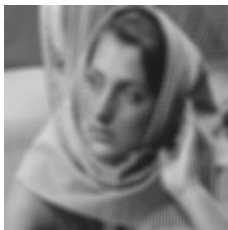
- It is an **ill-posed** inverse problem. We need add **prior information** on $\hat{u}$.

Example1: Image Deblurring (forward model)



Original image

$$\hat{u}$$



Blurred image

$$K\hat{u}$$



Degraded image

$$f = K\hat{u} + n$$

- The blurring effect is modelled by a known point spread function (PSF) k in the form of convolution:

$$f(x) = K\hat{u}(x) = k * \hat{u}(x) = \int_{\Omega} k(x - y)\hat{u}(y)dy.$$

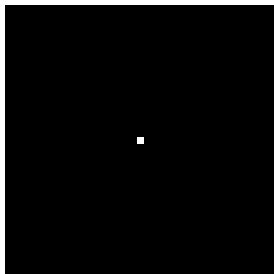
- The blurring operator K is a bounded linear operator and shift-invariant.

Example 1: Image Deblurring (point spread function)

To precisely describe the blurring operation, we need the point spread function (PSF). The PSF prescribes how the light from a single point in the sharp image is spread out in the blurred image. It satisfies $\int_{\mathbb{R}^2} k(x) \, dx = 1$.

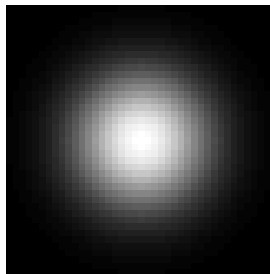
Point source

A single white pixel



Point spread function

Image of point source



Example1: Image Deblurring (inverse model)



Original image
 $\hat{u}$



Degraded image
 $f = K\hat{u} + n$

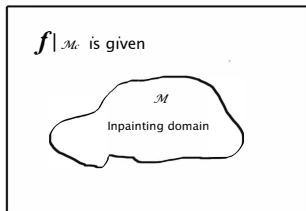
- Estimate $\hat{u}$ from f with n unknown
- It is a highly ill-posed inverse problem

Example2: Image Inpainting

The word “**inpainting**” is an artistic synonym for “image interpolation”.

Mathematical illustration: The image inpainting problem is to restore the original image $\hat{u}$ from the damaged image f , which is missing in inpainting domain $\mathcal{M} \subset \Omega$ and noisy in the rest part, i.e.

$$f(x) = \begin{cases} \hat{u}(x) + \epsilon, & x \in \Omega/\mathcal{M} \\ 0, & x \in \mathcal{M} \end{cases}$$



Example2: Image Inpainting

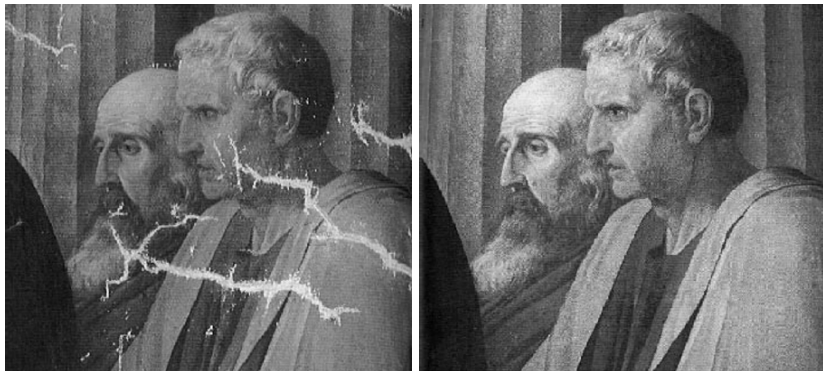


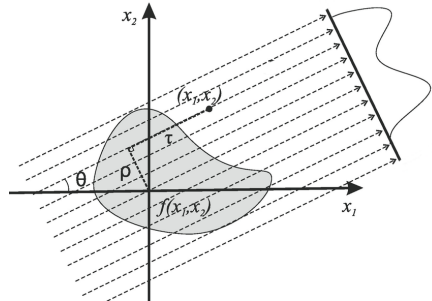
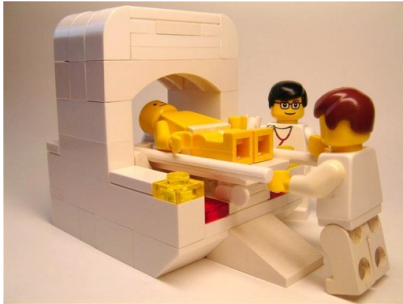
Figure: Resurrecting the art of the great masters.

Example2: Image Inpainting



Figure: Restoration of old photos.

Example3: Computed X-ray tomography (CT)



The principle in parallel-beam X-ray tomography: send parallel X-rays through the object at different angles, and measure the damping.

Variational Methods

- Variational methods play an important role in inverse problems.
- The variational method to imaging science constitutes the computation of a reconstructed image u based on the observed image (or more generally data) f as a minimizer of a functional, which is called as a variational model.
- The modelling of the variational method can be best motivated from Bayesian statistics.
- Variational methods can be used for denoising, deblurring, reconstruction, inpainting, super-resolution reconstruction, segmentation, registration, tomographic reconstruction, etc.

Variational Methods

Two main challenges:

① Good variational model: $\min_u E(u)$.

② Efficient optimization algorithm.

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 - Good image = small value of $E(\cdot)$
 - Existence, uniqueness, etc..
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- ② Efficient optimization algorithm.
 - The goal is to find an easily implemented algorithm to solve the variational model with low computation and high accuracy.
 - Convexity, relaxation, convergence, parallaxation, etc..

Variational Methods

From a statistical perspective, we wish to determine the most possible u from given measured data f , i.e.,

$$\max_u \pi(U = u|F = f).$$

Apply Bayes' Law

$$\pi(U = u|F = f) = \frac{\pi(F = f|U = u)\pi(U = u)}{\pi(F = f)},$$

then it is equivalent as

$$\max_u \pi(F = f|U = u)\pi(U = u) \iff \min_u -\log \pi(F = f|U = u) - \log \pi(U = u)$$

Variational Methods

$$\min_u -\log \pi(F = f|U = u) - \log \pi(U = u) \quad (*)$$

- With the additive Gaussian noise model, we have

$$\pi(F(x) = f(x)|U(x) = u(x)) = \mathcal{N}_{0,\sigma^2}(F(x) - Ku(x)),$$

$$-\log \pi(F = f|U = u) = \frac{1}{2\sigma^2} \int_{\Omega} [F(x) - Ku(x)]^2 dx.$$

- Based on prior information of U , we are able to choose the prior probability density of U as $\pi(U = u) = e^{-\tilde{\alpha}R(u)}$.

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Then (*) is equivalent to

$$\min_u \frac{1}{2\sigma^2} \int_{\Omega} [f(x) - Ku(x)]^2 dx + \tilde{\alpha}R(u).$$

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$$\min_u \frac{1}{2} \|f - Ku\|^2 + \lambda R(u).$$

Variational Methods

$$\min_u \frac{1}{2} \|f - Ku\|^2 + \lambda R(u).$$

- The first term is called as **data-fitting** or **data-fidelity** term, which depends on the noise model.
- The second term is called as **regularization** term, which depends on the prior information or assumption of the original image $\hat{u}$.
- $\lambda > 0$ is **regularization parameter**.

Review: Tikhonov Regularization

Tikhonov regularization

$$R_{\text{Tik}}(u) = \frac{1}{2}\|u\|^2$$

- **Idea:** We control the norm of the solution in order to suppress most of the large noise components.
- The variational model becomes

$$\min_u \frac{1}{2}\|f - Ku\|^2 + \frac{\lambda}{2}\|u\|^2.$$

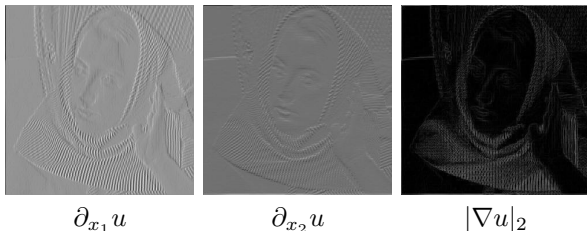
- The corresponding minimization problem is strictly convex.

Generalized Tikhonov Regularization

Generalized Tikhonov regularization

$$R_{\text{Tik}}(u) = \frac{1}{2} \|\nabla u\|_2^2$$

- **Idea:** We penalize large intensity changes.



- The variational model becomes

$$\min_u \frac{1}{2} \|f - Ku\|^2 + \frac{\lambda}{2} \|\nabla u\|_2^2.$$

- The corresponding minimization problem is strictly convex if and only if $\mathcal{N}(K) \cap \mathcal{N}(\nabla) = \{0\}$.

Total Variation

Total Variation for $u \in W^{1,1}(\Omega)$

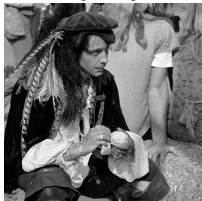
$$R_{\text{TV}}(u) = \|\nabla u\|_1$$

- **Idea:** We penalize intensity changes but allow outliers on ∇u .
- TV leads to stair-casing artifacts.
- TV is rotation invariant.
- The variational model becomes

$$\min_u \frac{1}{2} \|f - Ku\|^2 + \lambda \|\nabla u\|_1.$$

- The corresponding minimization problem is convex.

Original image



Noisy and blurred image



TV deblurred image



Total Generalized Variation

Total Generalized Variation for $u \in W^{2,1}(\Omega)$

$$R_{\text{TGV}}(u) = \min_v \beta_1 \|\nabla u - v\|_1 + \beta_2 \left\| \frac{1}{2}(\nabla v + \nabla v^T) \right\|_1$$

- **Idea:** We penalize higher order derivatives to avoid stair-casing artifacts.
- TGV is rotation invariant.
- The variational model becomes

$$\min_u \frac{1}{2} \|f - Ku\|^2 + \lambda R_{\text{TGV}}(u).$$

- The corresponding minimization problem is convex.

 f 

TV



TGV

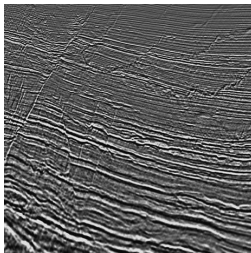
Directional Data

Data varies much less along one angular direction than the orthogonal direction(s).

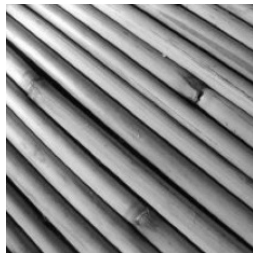
Examples



Fibers: optical, carbon,
glass(wind-turbine blades),
ceramic (fuel cells).



Seismic imaging



Digital imaging

Directional Total Variation

Directional Total Variation for $u \in W^{1,1}(\Omega)$

$$R_{\text{DTV}}(u) = \|\Lambda_a R_\theta \nabla u\|_1$$

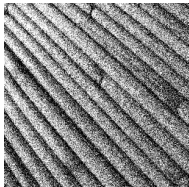
- **Idea:** We introduce the scaling and rotation to the gradient operator.
- Rotation matrix: $R_\theta = \begin{bmatrix} \cos \theta & -\sin \theta \\ \sin \theta & \cos \theta \end{bmatrix}$ with $\theta \in (-\frac{\pi}{2}, \frac{\pi}{2}]$
- Scaling matrix: $\Lambda_a = \begin{bmatrix} 1 & 0 \\ 0 & a \end{bmatrix}$ with $a \in (0, 1]$.
- Directional total variation is rotation variant.
- The variational model becomes

$$\min_u \frac{1}{2} \|f - Ku\|^2 + \lambda \|\Lambda_a R_\theta \nabla u\|_1.$$

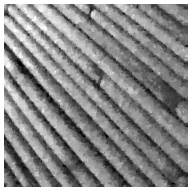
- The corresponding minimization problem is convex.

Image Deblurring

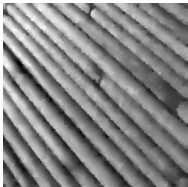
Noisy, psnr = 14.0



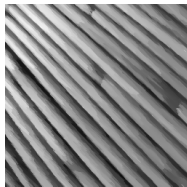
TV, psnr = 23.8



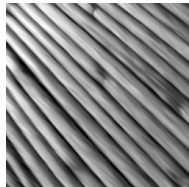
TGV, psnr = 24.7



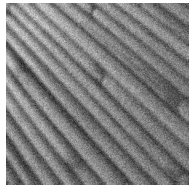
DTV, psnr = 26.8



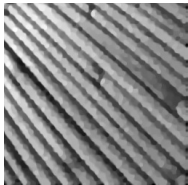
DTGV, psnr = 28.2



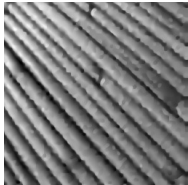
Degraded, psnr = 19.0



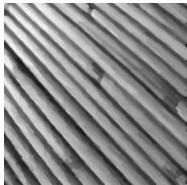
TV, psnr = 25.0



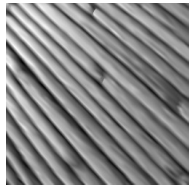
TGV, psnr = 25.5



DTV, psnr = 26.7



DTGV, psnr = 26.9



More Choices of Regularization Terms (Model-Based)

- Other Banach-space norms, e.g. Sobolev norms, nuclear norm, Besov norms, ...
- Anisotropic regularization
- Higher-order regularization; higher-order PDEs; Euler elastic; fractional regularization; ...
- Non-local regularization (non-local TV; non-local means, ...)
- Free-discontinuity problems, e.g. Mumford-Shah; ...
- Framelet based regularization, e.g. wavelets, curvelets, ...
- Mixtures of the above ...
- Many more ...

Sparse Coding and Dictionary Learning

- Sparse coding is a process to find the sparse representation of u .
- Dictionary Φ is over-complete. The advantage of having an over-complete basis is that they are better able to capture structures and patterns in u .
- The aim of sparse coding is to find a set of basis $\{\phi_i\}$ in Φ such that we can represent u as a linear combination of these basis

$$u = \sum_{i=1}^k \alpha_i \phi_i.$$

- The variational model

$$\min_{\alpha} \left\| K \left(\sum_{i=1}^k \alpha_i \phi_i \right) - z \right\|^2 + \lambda \|\alpha\|_1.$$

Neural Network as “Regularization” (e.g. Plug-and-Play)

Idea: Introduce an auxiliary variable to split the data-fitting term and the regularization term.

$$\min_u \frac{1}{2} \|Ku - z\|^2 + \lambda \mathcal{R}(v), \quad \text{s. t. } v = u.$$

It can be solved by, for example,

$$\min_{u,v} \frac{1}{2} \|Ku - z\|^2 + \lambda \mathcal{R}(v) + \frac{\mu}{2} \|u - v\|^2.$$

Apply alternating minimization algorithm:

$$\begin{aligned} u_k &= \arg \min_u \|Ku - z\|^2 + \mu \|u - v_{k-1}\|^2 \\ v_k &= \arg \min_v \frac{\mu}{2\lambda} \|v - u_k\|^2 + \mathcal{R}(v) \end{aligned}$$

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$$v_k = \arg \min_v \frac{\mu}{2\lambda} \|v - u_k\|^2 + \mathcal{R}(v)$$

$$\leftarrow \text{Denoiser}(u_k, \sqrt{\lambda/\mu})$$

Conclusions

- The choice of regularization should be based on the prior information of the exact solution $\hat{u}$.

Thank you!